File

Name: P4-fit-A4
Description: KISSsoft example
Changed by: AAA on: 30.10.2026 at: 19:34:49

1 Overview

Calculation method: According to DIN 7190-1:2017, elastic range
with additions such as centrifugal force, micro sliding, mounting, etc.
for a horizontal or longitudinal interference fit

Diameter shaft (mm)	[DiA]/[DiI]	50.00	/10.00			
Diameter hub (mm)	[DaA]/[DaI]	125.00	/50.00			
Length of Interference fit (mm)	[l]	40.00				
Diameter of joint (mm)	[DF]	50.00				
with special allowances:						
Upper allowance Shaft (µm)	[AoI]	65.0				
Lower allowance Shaft (µm)	[AuI]	65.0				
Tolerance measure Shaft (µm)	[TI]	0.0				
Upper allowance Hub (µm)	[AoA]	0.0				
Lower allowance Hub (µm)	[AuA]	0.0				
Tolerance measure Hub (µm)	[TA]	0.0				
Largest interference (µm)	[Po]	65.0				
Smallest interference (µm)	[Pu]	65.0				
Nominal torque (Nm)	[T]	0.00				
Application factor	[KA]	1.00				
Service torque (Nm)	[Tb]	0.00				
Axial force (N)	[FA]	0.00				
Bending moment (Nm)	[Mb]	0.00				
Radial force (N)	[Fr]	0.00				
Circumferential force (N)	[Fu]	0.00				
Speed (1/min)	[n]	0.00				
Interference (µm)	[P]	65.0	(65.0	..	65.0	)
Embedding (µm)	[s]	3.84				
Effective interference at 20°C(68°F) (µm)	[Pw]	61.16	(61.2	..	61.2	)
Effective interference at working temp.. (µm)	[PwTh]	61.16	(61.2	..	61.2	)
Service temperature shaft (°C)	[ThB]	20.0				
Service temperature hub (°C)	[ThB]	20.0				
Pressure stress by						
- Interference (after mounting) (N/mm²)	[pM]	44.23	(44.23	..	44.23	)
- Interference (working) (N/mm²)	[p]	44.23	(44.23	..	44.23	)
- Bending moment (N/mm²)	[pb]	0.00				
- Radial force (N/mm²)	[pr]	0.00				
Coefficient of friction, longitudinal	[µa]	0.120				
Coefficient of friction, circumferential	[µu]	0.120				
Safety against sliding	[Sr]	> 1000				
Required safety against sliding	[SSr]	1.50				

Equivalent stress according to von Mises
$\sigma V = (\sigma R^2+\tau T^2-\sigma R \cdot \tau T)^{0.5}$

1.1 Shaft

Material, own Input		C45 (1)			
Young's modulus (N/mm²)	[El]	215000.00			
Poisson's ratio (-)	[vl]	0.30			
Density (kg/m³)	[ρ]	7830.00			
Coefficient of thermal expansion (10^-6/K)	[α]	11.50			
Tensile strength (wall thickness) (N/mm²)	[Rm]	650.00			
Yield point (wall thickness) (N/mm²)	[Rp]	300.00			
The shaft's material strength values are defined using the wall thickness as the raw diameter.					
Component raw diameter (wall thickness) (mm)	[deff]	21			
Surface class of roughness	N6 Rz=4.8 (Grinding)				
Surface roughness (µm)	[Rz]	4.80			
External diameter (mm)	[DiA]	50.00			
Diameter increase during operation (µm)	[ΔD]	-8.06	(-8.06	..	-8.06)
Diameter increase during operation (o/oo)	[ΔD]	-0.16	(-0.16	..	-0.16)
Equivalent stress outer diameter (N/mm²)	[σVa]	46.18	(46.18	..	46.18)
- Radial stress (N/mm²)	[σRa]	-44.23	(-44.23	..	-44.23)
- Tangential stress (N/mm²)	[σTa]	-47.91	(-47.91	..	-47.91)
- With outside load (N/mm²)	[σVaMF]	46.18	(46.18	..	46.18)
Inner diameter (mm)	[Dil]	10.00			
Diameter increase during operation (µm)	[ΔD]	-4.29	(-4.29	..	-4.29)
Diameter increase during operation (o/oo)	[ΔD]	-0.43	(-0.43	..	-0.43)
Equivalent stress inner diameter (N/mm²)	[σVi]	92.14	(92.14	..	92.14)
- Radial stress (N/mm²)	[σRi]	0.00	(0.00	..	0.00)
- Tangential stress (N/mm²)	[σTi]	-92.14	(-92.14	..	-92.14)
Safety against fracture	[SiRm]	7.05	(7.05	..	7.05)
Required safety against fracture	[SSi.Rm]	1.50			
Safety against yield point	[Si.Rp]	3.26	(3.26	..	3.26)
Required safety against yield point	[SSi.Rp]	1.20			

1.2 Hub

Material, own Input		C45 (1)			
Young's modulus (N/mm²)	[EA]	70000.00			
Poisson's ratio (-)	[vA]	0.30			
Density (kg/m³)	[ρ]	7830.00			
Coefficient of thermal expansion (10^-6/K)	[α]	11.50			
Tensile strength (wall thickness) (N/mm²)	[Rm]	700.00			
Yield point (wall thickness) (N/mm²)	[Rp]	150.00			
The hub material strength values are defined using the wall thickness as the raw diameter.					
Component raw diameter (wall thickness) (mm)	[deff]	40			
Surface class of roughness	N6 Rz=4.8 (Grinding)				
Surface roughness (µm)	[Rz]	4.80			
External diameter (mm)	[DaA]	125.00			
Diameter increase during operation (µm)	[ΔD]	30.09	(30.09	..	30.09)
Diameter increase during operation (o/oo)	[ΔD]	0.24	(0.24	..	0.24)
Equivalent stress outer diameter (N/mm²)	[σVa]	16.85	(16.85	..	16.85)
- Radial stress (N/mm²)	[σRa]	0.00	(0.00	..	0.00)
- Tangential stress (N/mm²)	[σTa]	16.85	(16.85	..	16.85)
Inner diameter (mm)	[Dal]	50.00			
Diameter increase during operation (µm)	[ΔD]	53.10	(53.10	..	53.10)
Diameter increase during operation (o/oo)	[ΔD]	1.06	(1.06	..	1.06)
Equivalent stress inner diameter (N/mm²)	[σVi]	91.58	(91.58	..	91.58)
- Radial stress (N/mm²)	[σRi]	-44.23	(-44.23	..	-44.23)
- Tangential stress (N/mm²)	[σTi]	61.08	(61.08	..	61.08)

- Radial stress with outside load (N/mm²)	[σRai]	-44.23	(-44.23	..	-44.23	)
- Tangential stress with outside load (N/mm²)	[σTai]	61.08	(61.08	..	61.08	)
- With outside load (N/mm²)	[σViMF]	91.58	(91.58	..	91.58	)
Safety against fracture	[Si.Rm]	7.64	(7.64	..	7.64	)
Required safety against fracture	[SSi.Rm]	1.50				
Safety against yield point	[Si.Re]	1.64	(1.64	..	1.64	)
Required safety against yield point	[SSi.Re]	1.20				

Mere elastic stress, no verification of elastic plastic interference fit according to DIN 7190-1.

2 Service / Mounting / Dismounting

Transverse-interference-fit:						
Mounting clearance (mm)	[PsTh]	0.050				
Temperature difference for mounting:						
Shaft temperature: (°C) Hub temperature: [ThA] (°C)						
Mount with shaft temperature 20 °C:						
Assembly temperature shaft (°C)	[θI]	20				
Assembly temperature hub (°C)	[θA]	220.000				
Mount with shaft temperature -150 °C:						
Assembly temperature shaft (°C)	[θI]	-150				
Assembly temperature hub (°C)	[θA]	94.348				
calculated using the coefficient of thermal expansion						
Sub-cooling of the shaft according to DIN 7190-1 (10^-6/K)	[αw]	8.50				
Longitudinal pressure fit:						
Assembly temperature shaft (°C)	[θM]	20.00				
Assembly temperature hub (°C)	[θM]	20.00				
Coefficient of friction, longitudinal	[μe=μa*	1.300]	0.16			
Press on (force) (kN)	[Fpress]	43.35	(43.35	..	43.35	)
Coefficient of friction, longitudinal	[μll=μa*1.600]	0.19				
Press out (force) (kN)	[Fpress]	53.35	(53.35	..	53.35	)
Coefficient of friction	[μ]	0.19				
Max. torque to avoid Micro sliding (Nm)	[Tlimit]	270.11	(270.11	..	270.11	)

Notice concerning the display: Number-1 (Number-2.. Number-3):

Number-1: Value calculated with the mean allowance

Number-2: Value of the smallest possible allowance

Number-3: Value of the largest possible allowance

Notice: All strains and stresses are calculated for the purely elastic case.

End of report (lines: 157)
